

Practical No. 5

Aim: To deploy and manage a web application using AWS Elastic Beanstalk and understand the Platform as a Service (PaaS) environment in AWS.

Theory:

Platform as a Service (PaaS) is a cloud computing model that provides a managed platform to build, deploy, and manage applications without handling the underlying infrastructure.

AWS Elastic Beanstalk is a fully managed PaaS service that automatically handles deployment, load balancing, auto-scaling, and health monitoring. Developers simply upload their code, and Elastic Beanstalk takes care of the rest.

Advantages of PaaS

- Reduces infrastructure management overhead for developers
- Faster deployment and scaling of applications
- Built-in high availability through load balancing and auto-scaling

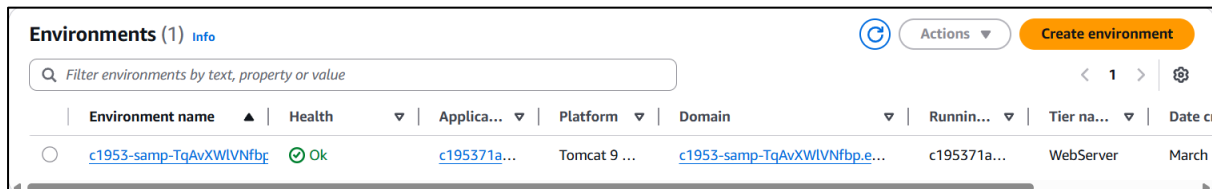
Disadvantages of PaaS

- Limited control over the underlying infrastructure
- Vendor lock-in to the AWS ecosystem
- Cost may increase significantly with high traffic and resource usage

Steps:

Task 1: Access the Elastic Beanstalk Environment

1. In the AWS Console, search for and open Elastic Beanstalk. The Environments page opens showing an existing application. Wait until the Health column displays Ok.



2. Click the environment name to open its Dashboard and verify the health status is Ok.
3. Click the Domain URL link at the top of the Dashboard. An HTTP 404 error is expected since no application is deployed yet.



Task 2: Deploy a Sample Application to Elastic Beanstalk

1. Download the sample Tomcat application from the AWS documentation (tomcat.zip).
2. In the Dashboard, click Upload and Deploy, select the tomcat.zip file, and click Deploy. Wait for the deployment to complete.

Upload and deploy

To deploy a previous version, go to the [Application versions page](#)

Upload application

Choose file

File name: tomcat.zip

File must be less than 500MB max file size

Version label

Unique name for this version of your application code.

c195371a5014304l14258192t1w530001336754-sampleapplicationversion-hoapidr

Deployment preferences

The application version will be deployed using the **AllAtOnce** policy.

Current number of EC2 instances: 2

Cancel

Deploy

- Once deployed, click the Domain URL to verify the web application is running successfully.

- Click Configuration in the left pane and review the Instance traffic and scaling panel for EC2 security groups, instance type, and scaling settings.
- Click Monitoring under Environment in the left panel and browse the available performance charts.

Task 3: Explore the AWS Resources Supporting the Application

- In the AWS Console, navigate to EC2 and click Instances. Two instances containing samp in their names are running, automatically created by Elastic Beanstalk.

Instances (2) Info									
Find Instance by attribute or tag (case-sensitive)									
All states									
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4	
☐	c1953-samp-T...	i-002372a3df35c09b7	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1c	ec2-3-81-4-	
☐	c1953-samp-T...	i-020b28cf3d9e31d1b	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	ec2-3-83-41	

- Explore the resources created by Elastic Beanstalk: a Security Group with port 80 open, a Load Balancer distributing traffic across both instances, and an Auto Scaling Group configured to run 2 to 6 instances based on load.

Conclusion: Thus, we have successfully deployed and managed a web application using AWS Elastic Beanstalk and understood the Platform as a Service (PaaS) environment in AWS.